Electric Aircraft Weight–Performance–Cost Estimation Method

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Purpose and calculation chain

The method sizes a battery-electric light aircraft from payload, MTOM (or target range) and pack specific energy, then derives mission performance and operating cost per hour, per km and per seat-km. It is a first-order conceptual tool, anchored to one reference airframe (Risen 912iS, 600 kg MTOM) and scaled by similarity laws. It is implemented in `risen_eas.py` and corresponds to EAS fidelity aero L0 / mass L1 / mission M0.

The calculation runs in five stages, each feeding the next:

1. **Inputs.** Payload (occupants + baggage), MTOM or required battery mass, pack specific energy E*, cruise speed and altitude, reserve policy, utilisation (h/yr) and mission mix.
2. **Geometry and mass closure.** Wing area, span and installed power scale with MTOM at constant wing loading and power-to-weight. Empty mass is built up by component, and the battery takes the remaining mass (fixed-MTOM mode) or MTOM is solved for a given battery and payload (solved-MTOM mode).
3. **Aerodynamics and power.** A parabolic drag polar gives drag and battery-side power at any speed, plus the characteristic speeds V_md (best range) and V_mp (best endurance).
4. **Mission performance.** Usable energy is split into sortie overhead (taxi, take-off, climb, descent), reserve and cruise. This gives mission range and endurance at beginning and end of pack life, plus C-rate and specific-power checks.
5. **Operating cost.** Capex, fixed annual costs and variable costs (electricity, per-pack battery wear, maintenance) are spread over a defined sortie and utilisation. A scaled piston (ICE) comparator uses the same airframe cost basis.

Inputs and default assumptions

The user supplies the design inputs in the first table; everything else defaults to the values in the second table. All defaults are overridable in code.

Design input	Symbol	Example (Option A)
Payload: occupants × 85 kg + baggage	W_p	180 kg (2 occ.) / 360 kg (4 occ.)

Design input	Symbol	Example (Option A)
MTOM (fixed-MTOM mode) or pack masses (solved-MTOM mode)	M, m_bat	274 kg main + 140 kg removable
Pack specific energy, pack level	E*	300 Wh/kg
Cruise speed / altitude	V_cr, h	60 m/s (216 km/h) / 1,500 m ISA
Reserve	t_res	30 min at V_md
Utilisation and mission mix	U, f_i	1,200 h/yr, 50/50 by hours

Group	Parameter	Default	Basis
Reference airframe	MTOM, S, b	600 kg, 9.70 m², 8.99 m	Risen 912iS datasheet
Reference airframe	Structure / fixed equipment / propeller	190 / 45 / 15 kg	Estimate from 325 kg empty mass less ~75 kg Rotax installation
Aerodynamics	C_D0 (electric)	0.0120	ICE-calibrated 0.0128 less 6.6 % cooling/exhaust drag
Aerodynamics	e, AR	0.80, 8.33	Estimate; geometry
Powertrain	Shaft power / mass at reference	75 kW / 30 kg (2.5 kW/kg system)	Rotax 912iS replacement
Efficiencies	η_prop cruise / climb	0.80 / 0.75	—
Efficiencies	η_motor · η_inverter · η_wiring · η_battery	0.95 · 0.98 · 0.99 · 0.96	Chain η = 0.708
Battery	Usable depth of discharge / end-of-life capacity	0.90 / 0.80	—
Battery	Cycle life / calendar life	1,500 cycles / 8 yr	To 80 % capacity
Charging	Cell × charger efficiency	0.96 × 0.92	Grid to pack

Group	Parameter	Default	Basis
Sortie	Taxi / take-off / climb power	3 kW × s, 5 min / full power, 1 min / 65/75 power at 40 m/s	s = M/600
Sortie	Descent	Glide at 55 m/s, 2 kW systems	No regeneration
Cost	FX / landed factor	₹110/€ / ×1.20	Aug 2026 rate
Cost	Airframe ex-engine / e-powertrain price	€229.5k × s^0.75 / €40k × s	Risen 2025 list €249.5k less €20k engine
Cost	Pack price / electricity tariff	₹70,000/kWh / ₹12/kWh	—
Cost	Residual / cost of capital / insurance	30 % / 9 % / 2.5 % of hull	Life 15 yr (<1,000 h/yr), 10 yr (≥1,000 h/yr)
Cost	Hangar + annual inspection	₹5.6 L/yr × h_f	h_f = 1 + 0.6(s − 1)
Cost	Maintenance reserves	₹350·s + 300·s^0.4 + 800·h_f per hour	Motor, propeller, airframe
Cost	DC charger / modular-pack kit	₹20 L × s (at ≥600 h/yr) / ₹5 L	8-yr amortisation

Weight estimation

Geometry scales at constant wing loading (61.9 kg/m²) and constant aspect ratio, so stall speed, V_md and V_mp do not change with size. Installed power scales with MTOM. With s = M / 600:

$$S = 9.70\,s,\qquad b=\sqrt{AR\,S},\qquad P_{sh}=75\,s\ \text{kW}$$

Structural mass uses the EAS L1 scaled-fraction law, with a wing share α = 0.5 and a wing-area exponent of 1.2. The super-linear wing term makes empty-mass fraction grow with size:

$$m_{struct}=190\left[\alpha\left(\frac{S}{S_0}\right)^{1.2}+(1-\alpha)\frac{M}{M_0}\right]$$

The other components scale as follows:

- **Fixed equipment** is 45 kg, plus 3 kg BRS growth above 600 kg and above 800 kg, plus 10 kg for a four-seat cabin.
- **Propeller** is 15 · s^0.7 kg.
- **E-powertrain** is 30 · s kg, i.e. a constant 2.5 kW/kg at system level.

The operating empty mass excluding battery is their sum:

```
OEW = m_{struct}+m_{fixed}+m_{prop}+m_{pt}
```

Mass closes in one of two modes.

1. **Fixed-MTOM mode** (the 600 kg and 750 kg two-seaters). The battery takes the mass that remains, and nominal energy follows from pack specific energy: m_bat = M − OEW − W_p, and E_nom = m_bat · E*.
2. **Solved-MTOM mode** (Option A). Each loading case i has a payload and a set of packs fitted. MTOM is the smallest M satisfying the heaviest case, found by bisection:

```
M = OEW(M) + \max_i\left(W_{p,i} + \sum_{j\in i} m_{bat,j}\right)
```

In Option A, the binding case is 4 occupants with the main pack only (360 + 274 kg). It gives M = 1,177.9 kg; the 2-occupant case with both packs then flies 40 kg under MTOM.

Performance estimation

Drag polar and power. A parabolic polar gives drag at any speed. Battery-side power divides thrust power by the full chain efficiency (η = 0.708 in cruise):

```
C_L=\frac{2W}{\rho V^2 S},\quad D=\tfrac12\rho V^2 S\,(C_{D_0}+kC_L^2),\quad
k=\frac{1}{\pi\,AR\,e},\quad P_{bat}=\frac{DV}{\eta}
```

Characteristic speeds. V_md is the minimum-drag (best-range) speed and V_mp the minimum-power (best-endurance) speed:

```
V_{md}=\sqrt{\frac{2W}{\rho S}}\left(\frac{k}{C_{D_0}}\right)^{1/4},\qquad
V_{mp}=\frac{V_{md}}{3^{1/4}},\qquad \left(\frac{L}{D}\right)_{max}=\frac{1}
{2\sqrt{C_{D_0}k}}
```

Ideal range and endurance. Battery mass is constant in flight, so the electric range equation is linear in energy. Usable energy is E_use = E_nom · DoD · f_EOL:

```
R_{ideal}=\frac{E_{use}\,\eta}{W}\left(\frac{L}{D}\right)_{max},\qquad
T_{ideal}=\frac{E_{use}\,\eta}{P_{min}},\qquad P_{min}=\frac{W}
{(C_L^{3/2}/C_D)_{max}}\sqrt{\frac{2W}{\rho S}}
```

Sortie overhead. Each sortie spends energy on four phases before and after cruise:

- **Taxi:** 3 kW × s for 5 min.
- **Take-off:** full shaft power for 1 min, divided by the electrical chain efficiency η_elec = 0.885.
- **Climb to 1,500 m:** at 65/75 power and 40 m/s. Rate of climb comes from excess power, ROC = (η_p,climb · P_cl − D·V) / W.
- **Descent:** a glide at 55 m/s with 2 kW for systems. Its ground distance is h · (L/D) at 55 m/s.

Mission range and endurance. The reserve E_res (30 min at V_md power) is held back. The remaining energy funds cruise:

```
R=d_{climb}+\frac{E_{use}-E_{ovh}-E_{res}}{P_{bat}
(V_{cr})}\,V_{cr}+d_{descent}
```

Endurance is the airborne time of the same calculation at V_mp. Both are evaluated at beginning of life (f_EOL = 1.0) and end of life (0.8).

Power checks. The take-off C-rate is the peak battery power divided by nominal energy, C_TO = (P_sh / η_elec) / E_nom. Pack specific power is P_sh / (η_elec · m_bat). These two checks bound cell selection: above about 2C or 600 W/kg, the 300 Wh/kg assumption is at risk.

Cost estimation

All costs are full-recovery operating costs with no operator margin. They are summed annually and divided by flight hours, km flown and occupied seat-km.

Capex. Airframe price scales with mass to the power 0.75, and the e-powertrain with power. A four-seat cabin adds 5 %:

```
C_{air}=\left(229{,}500\,s^{0.75}+40{,}000\,s\right)\times 110\times 1.2\
\text{INR},\qquad C_{pack,j}=E_{nom,j}\times 70{,}000
```

Fixed costs per year are the sum of the following items, where L is airframe life (15 yr below 1,000 h/yr, 10 yr at or above), r = 0.30 is residual value and i = 0.09 is cost of capital:

```
F=\frac{C_{air}(1-r)}{L}+i\left[\frac{C_{air}(1+r)}{2}+\frac{\sum C_{pack}}
{2}\right]+0.025\left(C_{air}+\sum C_{pack}\right)+5.6\text{ L}\cdot
h_f+\text{infra}
```

Infrastructure covers a DC charger (₹20 L × s, fitted at 600 h/yr or more) plus, for modular packs, a ₹5 L handling kit. Both are amortised over 8 years with capital charged on half the value.

Sortie model. Each operating mode i has a sortie length t_i (1.0 h for two occupants; 0.7 h for Option A four-up, so that the mission closes at end of life with reserve). The mode's share of hours f_i gives the number of sorties, n_i = U · f_i / t_i. Sortie energy E_s and distance d_s come from the performance model at 216 km/h.

Variable costs.

- **Electricity:** Σ n_i E_s,i / (0.96 × 0.92) × ₹12/kWh.
- **Maintenance:** (350·s + 300·s^0.4 + 800·h_f) ₹/h.
- **Battery wear, per pack.** Paralleled packs share energy throughput in proportion to capacity. Each pack's equivalent full cycles per year and life are:

$$\text{EFC}_j=\frac{E_{thr,j}}{0.9\,E_{nom,j}},\qquad L_j=\min\left(\frac{N_{cyc}}{\text{EFC}_j},\ 8\ \text{yr}\right),\qquad C_{bat}=\sum_j\frac{C_{pack,j}}{L_j}$$

The battery is therefore calendar-limited at low utilisation and cycle-limited at high utilisation. A pack that never cycles, such as Option A's removable pack in a four-up-only mix, still ages over 8 years.

ICE comparator. The same airframe cost law is used, with an engine in place of the e-powertrain:

MTOM	Engine price	Overhaul reserve
600 kg	€20k	₹1,375/h
750 kg	€45k	₹2,000/h
1,178 kg	€75k	₹3,300/h

Fuel is 8.57 L/h × s (datasheet long-range burn plus 1 L/h taxi/climb) at ₹120/L mogas, or ₹200/L avgas as a sensitivity. Airframe maintenance is ₹1,500/h × h_f. Distance is flown at 220 km/h.

Validation, limitations and reference implementation

The polar is validated against the Risen 912iS datasheet, and the code reproduces all 61 published configuration metrics within tolerance.

Aerodynamic calibration. Matching 73.5 kW shaft power at 175 kt TAS (sea level, 600 kg, η_p 0.80) gives an ICE C_D0 of 0.01285. That polar predicts 56.5 kW at the independent 75 %-power point (159 kt), against 55.1 kW on the datasheet (+2.5 %). The electric C_D0 of 0.0120 removes 6.6 % for engine cooling and exhaust drag, which raises (L/D)max from 20.2 to 20.9.

Reference implementation. `risen_eas.py` (Python 3, standard library only) is organised in six sections: inputs, mass closure, performance, cost, configurations, and verification. Running `python3 risen_eas.py --verify` checks every published metric against a relative tolerance: 0.2–0.5 % for masses, 1–1.5 % for ranges, endurances and costs, and 2–5 % for calibration points. It exits non-zero on any failure.

Limitations.

- **Fidelity is first order:** a fixed parabolic polar, constant propeller efficiency, constant pack specific energy, no SOC-dependent voltage sag and no regeneration.
- **Structural mass is the largest uncertainty.** ±10 kg of OEW is roughly ±3 kWh of battery, or ±20 km of range at 600 kg.
- **Cost coefficients are estimates, not quotes.** These include price scaling at M^0.75, maintenance rates, the insurance rate and battery cycle life. Cycle life alone moves cost per hour by about ±9 % across 1,000–2,500 cycles.
- **Certification is out of scope,** including the 600 kg ultralight limit and the reserve rules that apply to each category.